

MCA – 212 Object Oriented Software Engineering

Assignment – 1

1. Smart Hospital Management System

- Develop an integrated hospital management system for patient registration, appointments, doctor allocation, laboratory reports, pharmacy, billing, and electronic medical records.
- Include an AI-based module for **early risk identification or appointment prioritization**.
- Model interactions among patients, doctors, nurses, laboratory staff, pharmacists, administrators, and AI services.

2. Intelligent University Examination Management System

- Develop a system for examination scheduling, student eligibility, question-paper management, invigilation, evaluation, result processing, and revaluation.
- Use AI/ML to **identify potentially anomalous evaluation patterns or predict student performance**.
- Include role-based access for students, faculty, examination officers, and administrators.

3. Smart Agriculture Monitoring and Decision Support System

- Develop a system integrating soil sensors, weather information, crop records, irrigation schedules, and farmer activities.
- Use ML to **predict irrigation requirements or identify crop-health conditions** from sensor/image data.
- Provide dashboards and alerts for farmers and agricultural officers.

4. Intelligent Traffic Management System

- Develop a traffic management platform that collects data from cameras, sensors, traffic signals, and historical traffic records.
- Use AI for **traffic-density estimation and adaptive signal timing**.
- Support emergency vehicles, traffic authorities, citizens, and monitoring operators.

5. Smart Retail Inventory and Demand Management System

- Develop an inventory system managing products, suppliers, warehouses, sales, purchase orders, stock levels, and returns.
- Use ML for **demand forecasting and identification of abnormal inventory patterns.**
- Provide automated reorder recommendations while keeping final decisions with authorized users.

6. Intelligent Recruitment and Employee Management System

- Develop an HR platform for job posting, candidate registration, resume management, interview scheduling, employee onboarding, and performance records.
- Use NLP/ML to **extract candidate skills and rank profiles against job requirements.**
- Ensure that final recruitment decisions remain under authorized human control.

7. Smart Library and Knowledge Management System

- Develop a digital library supporting book acquisition, cataloguing, borrowing, returns, reservations, fines, digital resources, and user management.
- Use recommendation algorithms to **suggest books or research resources based on user interests and borrowing history.**
- Support students, faculty, librarians, administrators, and external users.

8. Intelligent E-Learning and Student Performance System

- Develop an educational platform supporting courses, assignments, quizzes, attendance, discussions, assessment, and academic progress.
- Use ML to **identify students requiring academic support and recommend learning resources.**
- Include separate interfaces for students, faculty, academic coordinators, and administrators.

9. Intelligent Logistics and Warehouse Management System

- Develop a logistics platform for order processing, warehouse allocation, inventory movement, shipment tracking, and delivery management.
- Use AI/ML for **delivery-time prediction and warehouse-demand forecasting.**

- Support suppliers, warehouse operators, logistics managers, delivery agents, and customers.

10. Smart Waste Management System

- Develop a municipal waste-management system for bin monitoring, collection scheduling, vehicle tracking, worker allocation, and complaint management.
- Use computer vision or ML for **waste-category identification and collection-demand prediction**.
- Provide dashboards for citizens, municipal officers, drivers, and supervisors.

11. Intelligent Cybersecurity Incident Management System

- Develop a security platform for collecting alerts, registering incidents, assigning analysts, tracking investigation activities, and generating reports.
- Use ML for **alert prioritization, anomaly detection, or incident classification**.
- Include authentication, role-based access control, audit trails, and incident escalation.

12. Smart Insurance Claim Processing System

- Develop a system for policy management, claim submission, document verification, surveyor assignment, claim assessment, approval, and settlement.
- Use AI for **document classification and identification of potentially suspicious claims**.
- Maintain human approval for important financial decisions.

13. Intelligent Disaster Management and Emergency Response System

- Develop a platform for disaster reporting, resource allocation, emergency-team coordination, shelter management, and communication.
- Use ML to **predict risk levels or prioritize emergency reports using multiple data sources**.
- Support citizens, emergency responders, government officers, hospitals, and relief organizations.

14. Smart Manufacturing and Predictive Maintenance System

- Develop an industrial system monitoring machines, production schedules, sensor readings, maintenance records, and spare parts.

- Use ML for **predictive maintenance and detection of abnormal machine behavior**.
- Support machine operators, maintenance engineers, production managers, and administrators.

15. Intelligent Food Delivery and Restaurant Management System

- Develop a platform connecting customers, restaurants, delivery partners, payment services, and administrators.
- Use ML for **delivery-time estimation, restaurant recommendations, or demand prediction**.
- Include order lifecycle management, payment processing, delivery tracking, and customer feedback.

16. Smart Tourism and Travel Management System

- Develop a travel platform supporting destination discovery, itinerary planning, hotel booking, transportation, attractions, reviews, and expense management.
- Use AI to **recommend destinations or generate personalized itinerary suggestions based on user preferences**.
- Support tourists, hotels, travel operators, local service providers, and administrators.

17. Intelligent Document and Government Service Management System

- Develop a digital government-service platform for citizen registration, application submission, document verification, workflow management, approvals, notifications, and service tracking.
- Use NLP/ML for **document classification, information extraction, and application prioritization**.
- Implement strong authentication, authorization, audit logging, data privacy, and multi-level approval workflows.